From Structural Centrality to Operational Resilience: Data-Driven Analysis and Recommendation in the Global Aviation Network

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***Abstract*—This study addresses a central tension in network science: whether structural centrality implies systemic vulnerability in real-world transportation systems. The so-called "Fragility Paradox" suggests that highly connected hubs, while topologically dominant, may represent points of failure due to traffic concentration and congestion dynamics.**

Despite extensive theoretical modeling of transportation networks, empirical validation of this hypothesis remains limited, particularly when combining global topology with large-scale operational performance data. Most studies either focus on static graph properties or analyze delay propagation in isolation, rarely integrating both perspectives within a unified analytical framework.

We therefore propose a structural-operational modeling approach that enables direct comparison between graph-theoretic importance and empirical delay behavior. By combining global route topology with large-scale flight records from multiple national aviation authorities, we evaluate whether topological dominance correlates with operational fragility or resilience.

Our findings indicate that systemic instability is primarily concentrated in peripheral nodes rather than in structurally dominant hubs, suggesting that resilience interventions should target local inefficiencies instead of central connectors.

I. Introduction

Air transport networks are commonly modeled as static graphs in which airports are nodes and routes are directed edges. Within this framework, structural centrality metrics such as PageRank and Betweenness Centrality are often interpreted as proxies for systemic importance and robustness. However, real-world aviation systems operate under dynamic constraints where delays, congestion, and resource limitations continuously reshape performance outcomes.

This study investigates the relationship between structural centrality and operational resilience in the global aviation network. Specifically, we examine whether topological importance translates into operational fragility, as suggested by the so-called "Fragility Paradox", which hypothesizes that highly central hubs may act as systemic bottlenecks due to traffic concentration.

To address this question, we construct an attributed aviation network integrating global route topology with millions of real flight records from the United States, United Kingdom, and Brazil. This unified representation enables a joint structural-operational analysis, allowing us to compare graph-theoretic importance with empirical delay behavior.

The analysis proceeds in four phases: (i) data engineering and network construction, (ii) topological characterization and robustness simulation, (iii) unsupervised clustering for airport health profiling, and (iv) development of a recommendation framework based on Locality Sensitive Hashing (LSH). The results provide empirical evidence that operational instability is concentrated in peripheral nodes rather than in structurally dominant hubs, challenging simplistic interpretations of centrality as fragility.

II. Phase 1 – Data Engineering and Truth Construction

The first phase focuses on constructing a coherent and analytically consistent representation of the global aviation network. The objective is to integrate structural topology with operational performance into a single attributed graph suitable for large-scale network analysis.

Rather than treating topology and performance as separate layers, we build a unified model in which airports are represented as nodes, direct routes as directed edges, and delay-related performance indicators as node attributes. This integration enables subsequent structural robustness tests and operational profiling within a common analytical framework.

A. Data Sources

The structural backbone of the network is derived from the OpenFlights database[1], which provides worldwide airport metadata and declared route connections.

Operational performance indicators were extracted from official 2024 flight records published by three national aviation authorities:

- **United States:** Bureau of Transportation Statistics (BTS)[2].
- **Brazil:** National Civil Aviation Agency (ANAC)[3].

[1]OpenFlights Data Portal
[2]BTS Data Portal
[3]ANAC Open Data Portal

- **United Kingdom:** Civil Aviation Authority (CAA)[4].

All operational datasets refer to the year 2024 and were harmonized to ensure consistency in delay definitions and aggregation logic.

B. From Raw Topology to a Consistent Network

The structural backbone is derived from the OpenFlights database, which initially provides **6,071** airports and **67,652** declared routes. Although globally comprehensive, the raw dataset requires substantial cleaning before it can support reliable network analysis.

Referential integrity checks remove **729** routes pointing to non-existent airports. A much larger issue concerns duplication: **29,881** parallel edges are merged, as OpenFlights records routes per *Airline*, generating multiple identical origin–destination connections for different carriers. Since these do not represent additional structural links, the network is modeled as a simple directed graph.

Finally, 2,814 isolated airports are pruned to retain only the connected backbone relevant for systemic analysis. The resulting graph consists of **3,257** active nodes and **37,041** unique directed edges, representing a structurally coherent and analytically robust connectivity network.

C. Extracting Operational Truth from Flight Records

Topology alone does not capture system performance. To evaluate fragility and efficiency, the structural graph is enriched with operational delay metrics derived from real flight records.

Flight data from three aviation authorities were processed: the United States (BTS), Brazil (ANAC), and the United Kingdom (CAA). To ensure methodological consistency, a strict **"Origin-Based" filtering logic** was applied. Since BTS and CAA data primarily track domestic/outbound departures, the ANAC dataset was filtered to exclude international flights arriving in Brazil (e.g., *Hong Kong* $\rightarrow$ *Sao Paulo*), retaining only flights originating from Brazilian airports.

The aggregation produces the following alignment:

- **USA (BTS):** 346 airports matched (96.4% coverage of raw data).
- **Brazil (ANAC):** 99 airports matched (81.8% coverage). This reduction reflects the focus on domestic operational efficiency.
- **UK (CAA):** 25 airports matched (100% coverage).

Overall, **470** airports receive delay indicators. As visualized in Figure 1, these nodes (colored points) represent the **operational layer** superimposed onto the global **structural backbone** (gray points). While this covers 14.4% of the total nodes, it provides a high-density, homogeneously defined sample of three major continental aviation systems.

[4]CAA Data Portal

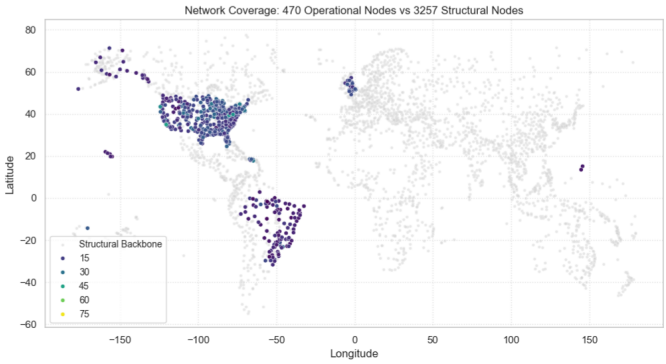

Fig. 1: **Network Coverage.** The map contrasts the global structural backbone (gray nodes, $N = 3257$) with the operationally enriched sub-network (colored nodes, $N = 470$). The operational layer focuses on the US, UK, and Brazil, ensuring high-quality delay metrics for these regions.

D. Bridging the Reality Gap and Network Augmentation

A validation step compares declared OpenFlights routes with routes actually observed in the flight records.

From real traffic data, **8,276** unique routes were extracted. Comparing these against the static backbone revealed that **3,223** active routes (38.9% of observed traffic) were missing from the OpenFlights database. This highlights a significant discrepancy between archival topology and dynamic operations.

Among the missing connections, **3,072** (95.3%) linked airports that already existed in the graph. These routes were injected into the network, increasing the edge count from 37,041 to **40,113**. The remaining unmatched routes were discarded as they involved nodes outside the connected giant component.

After augmentation, delay information covers 470 nodes and **8,125** edges (20.3% of the total).

E. Phase Outcome: Dual Graph Representation

At the end of Phase 1, two complementary network representations are available:

- A *structural graph* of 3,257 nodes and 40,113 empirically validated routes, used for global topological analysis.
- An *operationally enriched subgraph* defined over the 470 airports and 8,125 edges for which rigorous delay metrics are available.

F. Diagnostic Validation

To verify the consistency of the harmonized dataset, we analyzed the relationship between traffic volume, delay magnitude, and predictability.

Figure 2 highlights a distinct "funnel" pattern governed by the **Law of Large Numbers**. Low-volume airports exhibit naturally higher volatility, whereas high-volume hubs converge toward a stable delay band.

Crucially, the filtering of non-domestic Brazil flights has eliminated the **sampling bias** responsible for the extreme anomalies observed in preliminary tests. Previously, global hubs like Hong Kong were evaluated based solely on a

sparse subset of their operations—specifically, a few long-haul connections to Brazil. Using such a limited sample size effectively projected the performance of a single, complex route onto the entire airport profile, resulting in artificially inflated delays (> 350 minutes). By restricting the analysis to domestic departures where flight volumes ensure statistical significance, the maximum average delay has stabilized at a realistic ≈ 76 minutes.

Figure 3 confirms a cleaner linear correlation between Mean Delay and Variance. The disappearance of low-variance/high-delay outliers indicates that the network contains no deterministic bottlenecks; rather, inefficiency is consistently associated with operational instability (chaos), reinforcing the need for the robustness analysis in Phase 2.

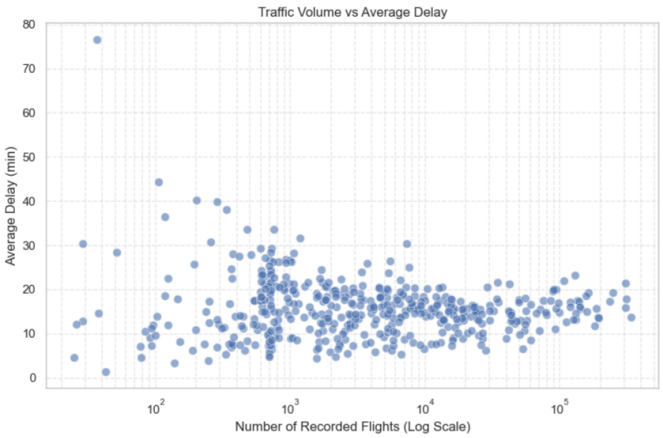

Fig. 2: **Volume vs Delay.** The removal of international inbound artifacts reveals a cleaner "funnel" effect, with max average delays stabilizing around 76 minutes.

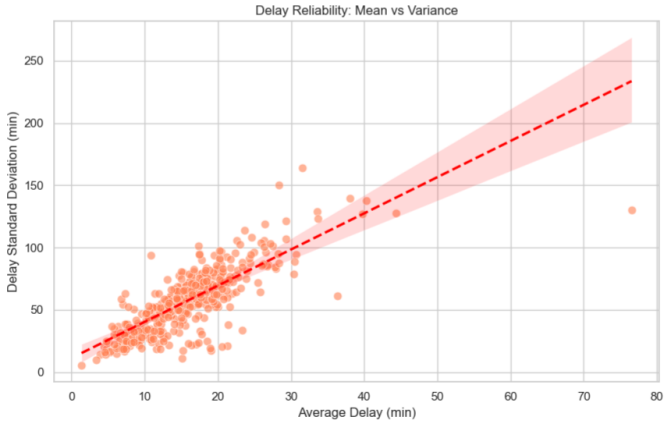

Fig. 3: **Reliability.** The corrected dataset shows a strong linear correlation: delay breeds unpredictability, without deterministic outliers.

III. Phase 2 – Topological Analysis and the Fragility Paradox

Having established the augmented graph in Phase 1, the second phase focuses on dissecting the network's anatomy. The objective is twofold: to characterize the global topology using graph theory metrics and to stress-test the system to verify the *Fragility Paradox* hypothesis.

A. Topological Characterization

The global aviation network follows specific organizational principles dictated by geography, politics, and airline economics. To capture this structure, we computed fundamental metrics on the *Giant Connected Component* (GCC), which covers 99.2% of the nodes (3,231 airports), ensuring our analysis represents the functional core of the system.

1) Scale-Free Property and Degree Distribution: A fundamental signature of real-world complex networks is the *Scale-Free* property, characterized by a degree distribution that follows a Power Law: $P(k) \sim k^{-\gamma}$. This implies a hierarchy dominated by a few "super-hubs" holding the network together.

To rigorously verify this, we applied three distinct methodologies in logarithmic scale (Figure 4):

- **Method A (Classic PDF):** The standard Probability Density Function reveals the typical heavy tail of empirical networks, though affected by statistical noise at the extremities ($\gamma \approx 1.05$).
- **Method B (Robust CCDF):** The Complementary Cumulative Distribution Function ($P(X \geq k)$) smooths the tail noise, providing a more reliable structural signature ($\gamma \approx 1.72$).
- **Method C (Rank-Degree Plot):** Following the Zipfian approach, we plotted the rank r against the degree k. The clear linear trajectory confirms the hierarchy ($\gamma \approx 1.78$).

The convergence of robust methods around $\gamma \approx 1.75$ confirms the network is Scale-Free. This "condensed" exponent (lower than the theoretical 2.0-3.0 range) is characteristic of physical transportation networks, where infrastructure costs limit the maximum number of connections a single node can physically handle.

2) Directed Assortativity: The directed assortativity coefficients computed for the four degree combinations $(out, in) = -0.0424$, $(out, out) = -0.0435$, $(in, in) = -0.0406$, and $(in, out) = -0.0418$ all lie in a very narrow interval around -0.04. This indicates a weakly disassortative but almost neutral mixing pattern in the global airport network.

The slightly negative values suggest that high-degree airports tend, on average, to connect to lower-degree airports, reflecting a typical hub–spoke organization. However, the small magnitude of the coefficients shows that this tendency is limited. The presence of direct connections among major hubs counterbalances the hub–periphery structure, resulting in an overall near-neutral assortative behavior. In other words, the worldwide airport network exhibits a mixed topology in which hub–spoke interactions dominate numerically, but are partially offset by inter-hub connectivity.

B. Centrality Landscapes: PageRank vs. Betweenness

Centrality captures different notions of importance within the network. We focus on two complementary metrics:

- **PageRank ($\alpha = 0.85$):** Measures how strongly an airport is connected to other well-connected airports, effectively identifying hubs that handle large volumes of traffic and are frequently reached by flights.

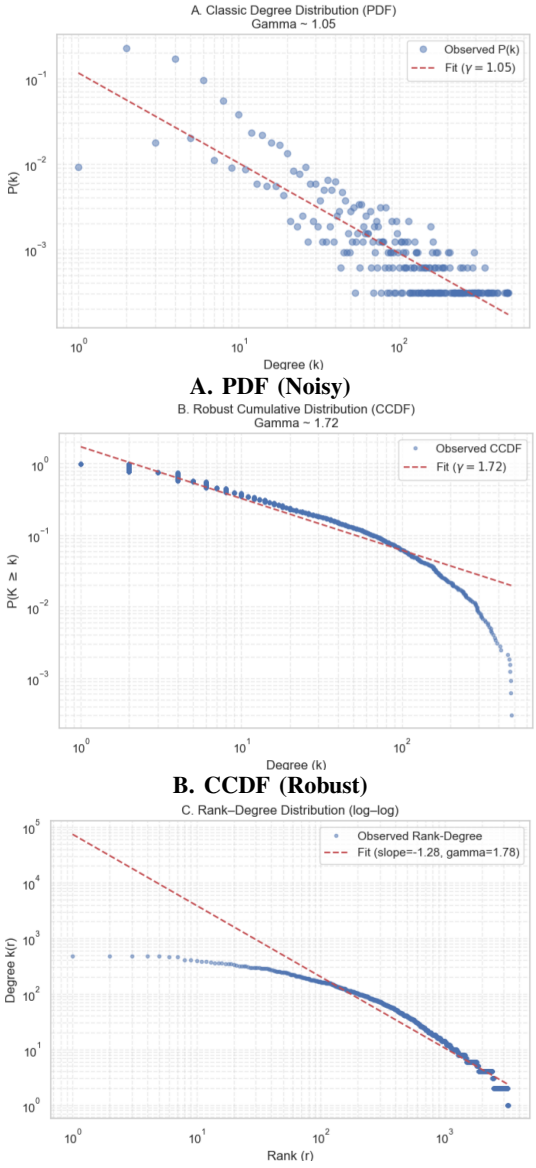

Fig. 4: **Scale-Free Validation Panel.** While the classic PDF (A) suffers from tail noise, the robust CCDF (B) and Rank-Degree (C) methods reveal a clear Power Law structure with $\gamma \approx 1.75$.

- **Betweenness Centrality:** Quantifies how often an airport lies on the shortest paths between other airports, highlighting strategic "bridge" nodes that control flow across the network.

Table I shows the functional divergence between these metrics. PageRank highlights high-traffic hubs (DFW, IST, ORD), while Betweenness identifies airports crucial for connecting regions globally: **Dubai (DXB)** links East and West, and **Anchorage (ANC)** serves as a key transpacific cargo and refueling node.

C. Testing the Fragility Paradox

The "Fragility Paradox" suggests that structurally central nodes should be more prone to congestion and delays. To test this, we analyzed the relationship between *Betweenness Centrality* and *Average Delay*.

Rank	PageRank (Volume Giants)	Rank	Betweenness (Bridge Giants)
1	DFW (Dallas)	1	DXB (Dubai)
2	IST (Istanbul)	2	CDG (Paris)
3	ORD (Chicago)	3	ANC (Anchorage)
4	ATL (Atlanta)	4	LAX (Los Angeles)
5	DEN (Denver)	5	FRA (Frankfurt)

TABLE I: Comparison of Top 5 Airports by Centrality Metric. Note the emergence of ANC and DXB as critical "bridges" in the Betweenness ranking.

Given the non-normal distribution of delays, we computed:

- **Pearson (r):** -0.0220
- **Spearman (ρ):** -0.1029 (p-value $= 0.026$)

Outcome: Hypothesis REJECTED. The correlation is statistically negligible, indicating no linear increase in delays for central nodes. However, the visual analysis reveals a significant difference in *variance*. As shown in Figure 5, peripheral nodes (low betweenness) exhibit extreme volatility, ranging from 0 to nearly 80 minutes of delay. In contrast, high-centrality nodes show stable, predictable performance with delays consistently clustered in a narrow lower range. Thus, centrality does not cause fragility; rather, it correlates with operational stability, likely due to better infrastructure and resource management at major hubs.

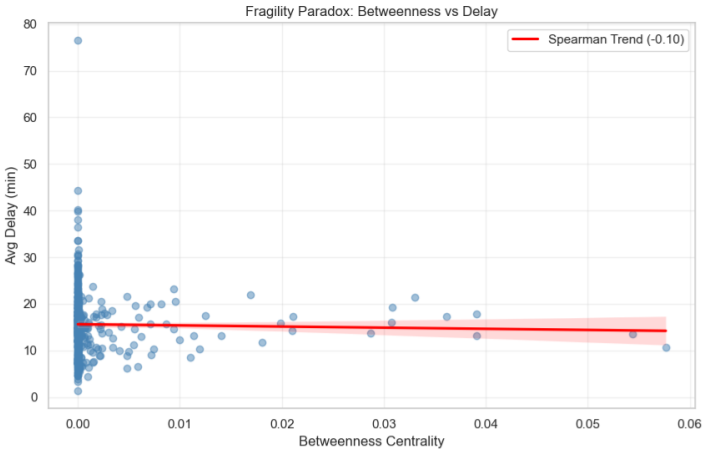

Fig. 5: **Rejection of the Fragility Paradox.** The linear trend is almost flat ($\rho \approx -0.10$). Central nodes have consistently low delays, while peripheral nodes show wide variation in delay times.

D. Robustness Simulation: The Crash Test

Finally, we simulated network disintegration by sequentially removing nodes based on specific strategies (Figure 6).

1) **Random Failure (Grey):** The network shows high resilience typical of scale-free topologies. Removing 20% of nodes randomly leads to a linear, gradual decline in connectivity rather than a sudden collapse.
2) **Structural Attack (Red):** Removing central nodes (targeted attack) is catastrophic. Eliminating just the top 10% of hubs collapses the Giant Component by $\approx 70\%$, shattering the network into disconnected islands.

3) **Operational Collapse (Orange):** *The crucial finding.* Removing the airports with the **highest delays** follows a trajectory almost identical to Random Failure.

This simulation confirms our earlier findings: the "sick" (high delay) nodes are **not** the structural pillars. If the Super-Hubs were the ones failing, the Orange curve would track the Red curve. Instead, the network remains topologically robust even if its most inefficient airports collapse. This proves that inefficiency is currently marginalized to the network periphery, sparing the backbone from operational contagion.

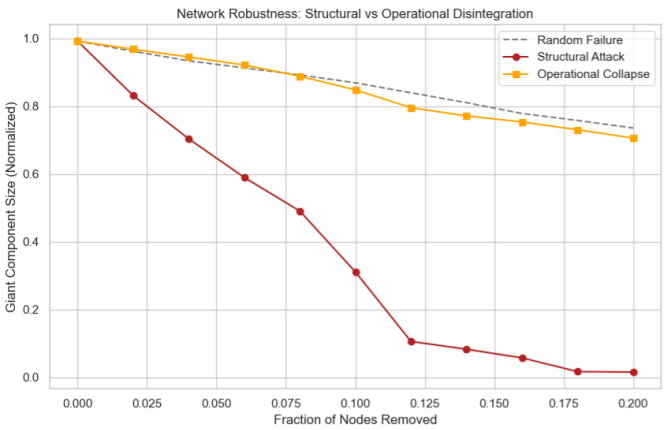

Fig. 6: **Network Robustness.** The Orange line (Operational Collapse) diverges significantly from the Red line (Structural Attack) and adheres to the Random Failure baseline. This demonstrates that high-delay nodes are structurally irrelevant to the global connectivity.

IV. Phase 3 – Health-Based Clustering and Profiling

While the correlation analysis in Phase 2 demonstrated general trends, the global aviation network is too complex to be defined by simple linear relationships. The objective of Phase 3 is to build a multidimensional data-driven taxonomy of "airport health."

By grouping airports based on their combined structural and operational profiles, we aim to isolate specific nodes that act as systemic bottlenecks, distinguishing them from efficient hubs and peripheral infrastructure. The analysis follows a sequential pipeline: exploratory hierarchical analysis, followed by partition validation and semantic profiling.

A. Preprocessing and Exploratory Hierarchical Structure

To capture both the topology and the operational reality of each node, we constructed a 5-dimensional feature space for the 470 airports with complete data:

- **Features:** *PageRank*, *Betweenness Centrality*, *Degree*.
- **Operational Features:** *Average Delay*, *Delay Variance*.

Since these features exist on vastly different scales, with PageRank values on the order of 10^{-4} and delays measured in hundreds of minutes, we applied a **Z-score Standardization** (`StandardScaler`). This ensures that variance in delay does not mathematically overwhelm topological importance during distance computations.

As an initial exploratory step, we applied **Hierarchical Clustering** (Ward's minimum variance method) to the standardized dataset. To maintain visual clarity, the dendrogram was truncated to display the last 30 merged branches (Figure 7).

The dendrogram reveals a clear multi-scale structure:

1) **High-level split ($y \approx 35$):** The network separates into two major groups of airports.
2) **Lower-level fragmentation ($y \approx 20$):** These groups further divide into approximately five clusters, indicating the presence of differentiated operational and structural profiles across airports.

Fig. 7: **Global Hierarchical Dendrogram.** Using Ward's linkage on all 470 nodes, the network shows two major groups at $y \approx 35$, which further subdivide into about five clusters at $y \approx 20$, providing a basis for partitioned clustering.

B. K-Means Segmentation and Validation

To segment the airport network, we ran a Grid Search using **K-Means** for $k \in [2, 6]$ and evaluated three internal validation metrics:

- **Silhouette Score (Sil):** measures how well each node fits within its cluster relative to other clusters. Higher values indicate well-separated, cohesive clusters.
- **Davies-Bouldin Index (DB):** quantifies cluster overlap; lower values indicate clusters that are more distinct and compact.
- **Calinski-Harabasz Index (CH):** ratio of between-cluster variance to within-cluster variance; higher values indicate dense, internally cohesive clusters.

The metric trends (Figure 8) reveal two potential choices:

- $k = 2$ **(Primary Dichotomy):** Silhouette (0.676) and Davies-Bouldin (0.679) clearly outperform the other k values. This indicates two clusters that are well-separated and fairly homogeneous internally, likely corresponding to a broad core/periphery division. The Calinski-Harabasz index is lower (304.7), suggesting that while clusters are separated, their internal cohesion is moderate.
- $k = 5$ **(Functional Granularity):** Silhouette (0.418) and Davies-Bouldin (0.764) are lower than $k = 2$, meaning clusters are closer together and less separated. However,

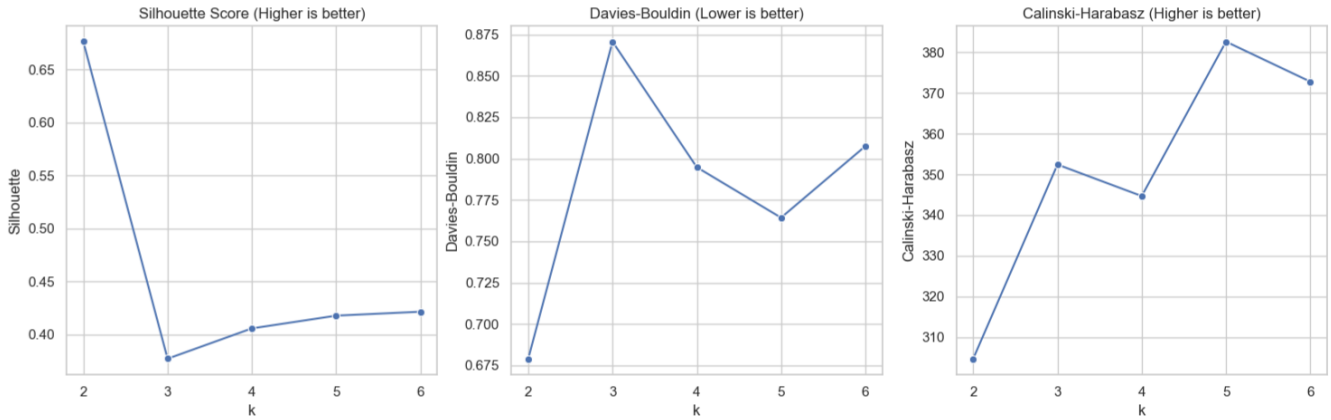

Fig. 8: **K-Means Validation Metrics.** Silhouette and Davies-Bouldin favor $k = 2$, indicating well-separated clusters, while Calinski-Harabasz peaks at $k = 5$, highlighting dense, cohesive groupings. Both k values merit further exploration for profiling.

the Calinski-Harabasz index peaks (382.6), implying that these five clusters are internally very dense and cohesive. This suggests that the network can be partitioned into finer operational "phenotypes" while still maintaining meaningful group structure.

Following the dendrogram insights, both **k=2** and **k=5** are valid candidates. Exploring both allows us to see whether a coarse core/periphery or a more detailed functional partition better captures airport health patterns in the network.

C. Outlier Detection (DBSCAN)

Partition-based algorithms like K-Means assign every data point to a centroid, which can distort clusters when anomalous nodes exist. To better understand the distribution of extreme observations, we employed **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise), which identifies clusters based on local density rather than global distance.

Theoretical Mechanism: DBSCAN does not require a predefined number of clusters. Each node is classified based on a radius ϵ and a minimum point threshold ($MinPts$):

- **Core Points:** Nodes in high-density regions with at least $MinPts$ neighbors within ϵ.
- **Border Points:** Nodes within the radius of a core point but not dense enough to be core themselves.
- **Noise (Outliers):** Nodes in sparse regions, unreachable from any core point.

For visualization and initial assessment, we projected the dataset into 2D using PCA. The first two components explain $\approx 53\%$ and $\approx 35\%$ of the variance, respectively.

Parameter Selection: The optimal radius ($\epsilon = 1.0$) was determined using the "elbow" in the k-Nearest Neighbor distance plot (Figure 9), which indicates the threshold where distances begin to increase sharply, separating dense regions from sparse points.

Results & Interpretation: DBSCAN identified 22 outliers ($\approx 4.7\%$ of the dataset) (Figure 10). These nodes lie on the fringes of the PCA projection and do not form dense clusters. While they are extreme points in the feature space, we do not discard them. In a variance-driven algorithm like K-Means, such nodes could be grouped into a distinct cluster, allowing us to investigate whether they represent a meaningful structural or operational profile rather than treating them as noise.

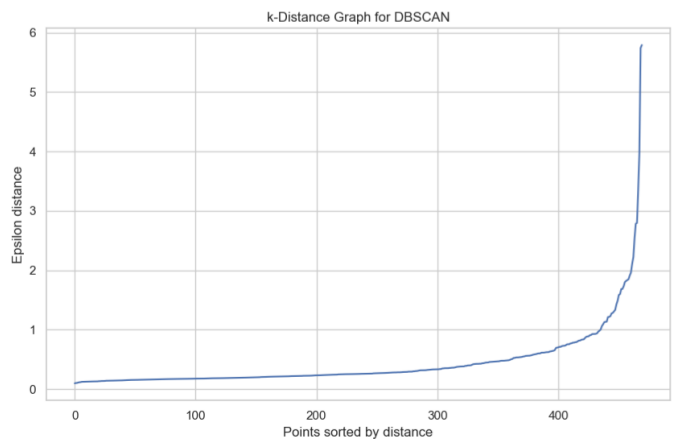

Fig. 9: k**-Distance Graph.** The "elbow" around $\epsilon = 1.0$ determines the DBSCAN radius.

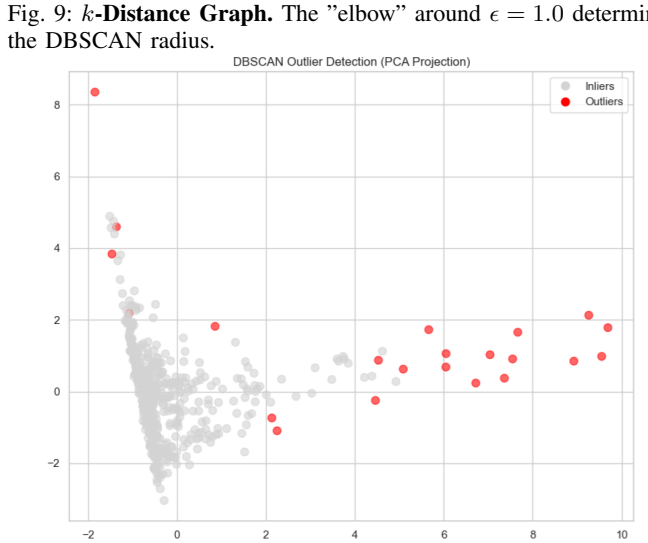

Fig. 10: **DBSCAN Outliers in PCA Space.** DBSCAN isolates 22 nodes (red) that are located on the fringes of the first two PCA components. These points are retained for subsequent K-Means analysis to evaluate whether they form a distinct cluster.

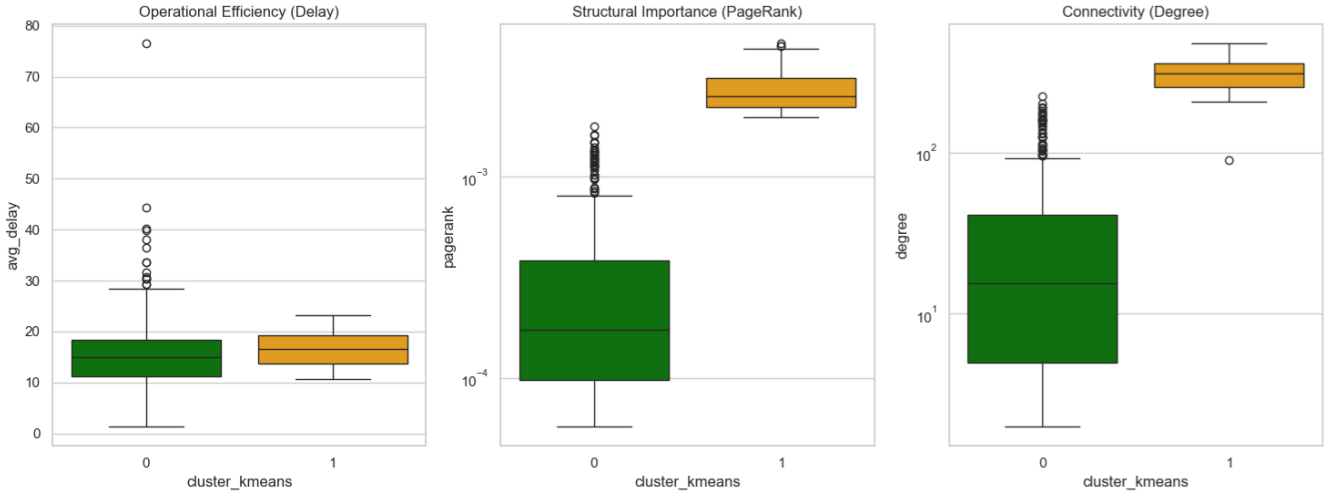

Fig. 11: **Feature Boxplots** ($k = 2$)**.** Note the lower median but higher variance in Cluster 0.

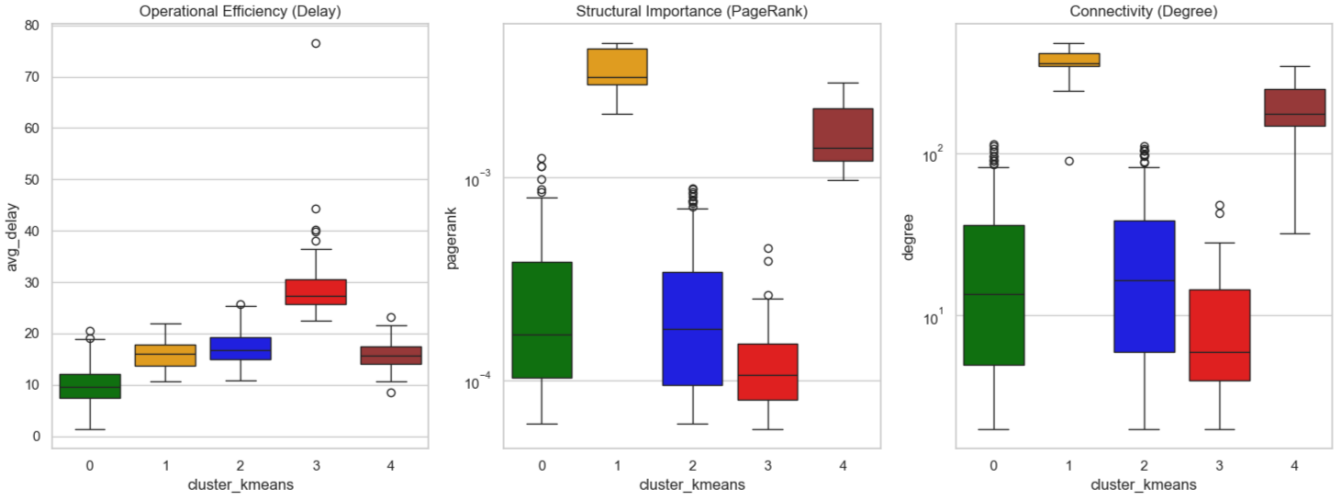

Fig. 12: **Semantic Profiling** ($k = 5$)**.** The Boxplots reveal the distinct behaviors. The **Orange** cluster (Center) has the highest PageRank but low Delays. The **Red** cluster (Right) shows the highest Delays despite low structural importance.

D. Macro-Structural Segmentation ($k = 2$)

As suggested by the Silhouette and Davies-Bouldin scores, the most mathematically distinct partition divides the network into two "Super-Groups". This confirms the scale-free nature of the aviation network, characterized by a stark "Core-Periphery" dichotomy.

Analyzing the feature distributions (Figure 11), we identify:

1) **Cluster 0 (Green) –** This group contains the vast majority of airports. They are characterized by low Degree and PageRank. Operationally, their performance is highly volatile: although the **median delay is slightly lower** than that of the backbone cluster, the variance is extreme.
2) **Cluster 1 (Orange) –** A small elite of high-degree nodes. Despite handling the highest traffic volume, these nodes maintain a stable operational profile with a higher median delay but a much more predictable behavior.

A crucial observation emerges when cross-referencing these results with the previous density analysis: the **majority of the 22 outliers identified by DBSCAN have been assigned to Cluster 0**.

E. Functional Health Profiling ($k = 5$)

To capture the operational details hidden by the binary split, we applied the partition favored by the Calinski-Harabasz Index ($k = 5$). The resulting profiles (Figure 12) reveal that "Fragility" is not evenly distributed across the periphery but concentrated in specific pockets.

Based on the boxplots, we derived the following semantic labels:

1) **Cluster 1 (Orange) –** The highest structural importance (Max PageRank and Degree). Crucially, they exhibit **low and stable delays**. This definitively rejects the Fragility Paradox: the most central nodes are arguably the most robust, leveraging economies of scale to buffer against congestion.

2) **Cluster 4 (Brown) –** High-degree nodes that act as secondary hubs. They mirror the Super-Hubs in efficiency, maintaining low delays despite significant traffic loads.
3) **Cluster 3 (Red) –** *The critical finding.* This cluster has low PageRank and Degree (similar to the general periphery) but exhibits the **highest average delay** and variance of the entire network. These are the true bottlenecks: structurally irrelevant nodes that suffer from systemic operational failures.
4) **Cluster 2 (Blue) –** Structurally, these nodes are nearly identical to Cluster 0 (low-to-medium connectivity). However, they exhibit notably higher average delays. They represent regional airports that, unlike their efficient peers, fail to maintain operational fluidity.
5) **Cluster 0 (Green) –** The best-performing group in the network. While structurally similar to Cluster 2, these nodes maintain the **lowest average delay** and variance. They represent the ideal operational standard for regional infrastructure.

Conclusion: The $k = 5$ segmentation proves that systemic risk is not located in the Core (Orange/Brown), but in a specific subset of inefficient peripheral nodes (Red).

Fig. 13: **PCA Projection** ($k = 2$)**.** The binary split.

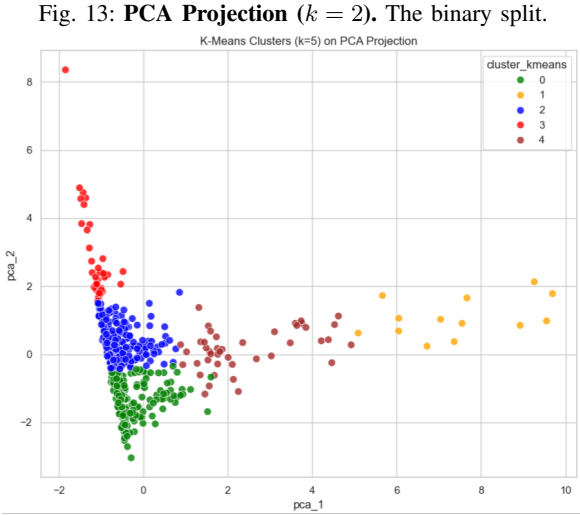

Fig. 14: **PCA Projection** ($k = 5$)**.** The network splits into distinct operational phenotypes.

V. Phase 4: Smart Switch Recommendation System

The final objective of this study is to move from descriptive analysis to a prescriptive solution. We developed the "Smart Switch" engine, a recommendation system designed to identify operational "twins": airports that share similar connectivity patterns but offer significantly better performance.

A. LSH Indexing: Didactic Demonstration vs. Necessity

For our dataset of approximately 470 nodes, a full *Brute Force* search for nearest neighbors is already computationally trivial (avg. 0.0029s). Using LSH here is therefore not strictly necessary; it was implemented primarily for didactic purposes and to illustrate a scalable approach applicable to much larger networks.

Conservative Parameterization: To avoid losing any potentially useful candidates (*False Negatives*), we opted for a cautious configuration. Using 128 permutations, 64 bands, and 2 rows, the theoretical threshold is:

$$t = \left(\frac{1}{b}\right)^{\frac{1}{r}} = \left(\frac{1}{64}\right)^{\frac{1}{2}} \approx 0.125$$

This low threshold favors high sensitivity, ensuring that virtually no valid alternative is missed. The resulting S-Curve (Figure 15) reflects this "greedy" retrieval strategy focused on **Recall**.

Correctness & Efficiency Benchmark: Even with this small dataset, LSH performs well compared to Brute Force:

- **Recall (91.84%):** Almost all true candidates are captured.
- **Accuracy (92.71%):** Overall reliability remains high.
- **Speedup (17x):** LSH is already an order of magnitude faster, demonstrating the efficiency of the MinHash approach; note that this value varies considerably between runs due to the limited dataset size.

This presentation emphasizes that LSH was used as an educational illustration of scalability, rather than a strict necessity for the problem at hand.

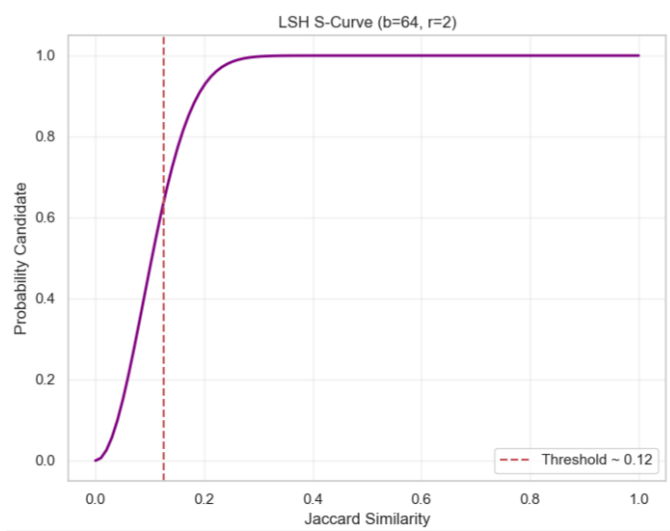

Fig. 15: **LSH S-Curve.** Configuration ($b = 64, r = 2$) with a threshold of 0.12, maximizing candidate retrieval.

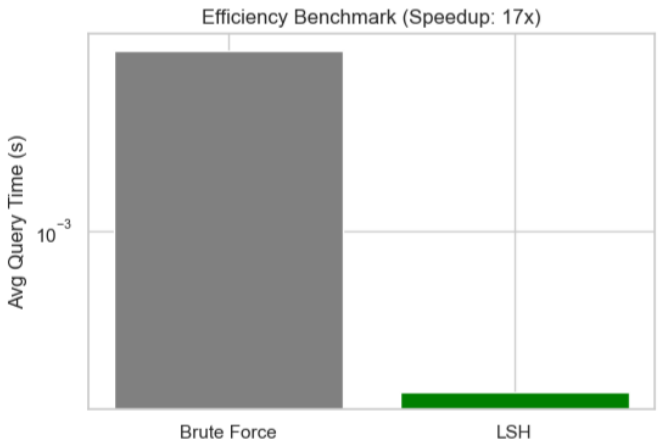

Fig. 16: **Efficiency Benchmark.** Comparison between Brute Force and LSH (Log Scale).

B. The "Smart Switch" Recommendation Logic

The engine targets airports within the **Bottleneck Cluster** (the cluster identified with the highest average delay). For each critical node, the system identifies and validates potential alternatives through a sequential multi-layer pipeline:

1) **Candidate Retrieval (LSH):** The system first queries the LSH index to extract a preliminary set of potential "twins" that share a theoretical topological resemblance with the bottleneck airport.
2) **Business Logic Filtering:** To reduce computational overhead and ensure feasibility, the candidates are immediately screened against two operational constraints:
   - **Capacity Constraint:** The candidate's *Degree* must be at least **50%** of the bottleneck's degree to ensure it can absorb the redirected traffic volume.
   - **Operational Gain:** A candidate is retained only if it offers a potential reduction in **Average Delay of at least 5 minutes**.
3) **Exact Similarity Validation:** For the candidates that pass the operational filters, the system calculates the exact **Jaccard Similarity**. A final recommendation is issued only if the similarity is > 0.4, guaranteeing that the recommended "Twin" covers a broad and meaningful portion of the original destinations.

By applying a **Smart Score** calculated as $Potential_Gain \times Similarity$, the system ranks the strategic alternatives, effectively prioritizing switches that offer the best balance between operational relief and topological compatibility.

C. Comparative Analysis: Results for $k = 2$ vs $k = 5$

The performance of the "Smart Switch" engine varies significantly depending on the granularity of the underlying clusters, revealing a trade-off between the *strategic relevance* of the nodes and the *magnitude of operational gain*.

1) Scenario A: Results for $k = 2$ (Backbone Optimization): In the binary partition, the system identifies Cluster 1 (the backbone hubs) as the bottleneck due to its higher median delay compared to the periphery. However, this target presents limited room for improvement:

- **Operational Scope:** The engine attempts to optimize 28 major hubs (e.g., DFW, MIA, CLT) with an average baseline delay of 16.6 min.
- **Efficiency:** Recommendations occur mostly within the same cluster (Hub-to-Hub). Due to the high-capacity constraint ($> 50\%$ degree), only 74 total candidates are found.
- **Impact:** The average potential gain is a modest **6.5 minutes**. The system suggests switching between mega-hubs (e.g., DFW to ATL), providing high-quality matches (Jaccard > 0.4 for the top 20) but minimal relief. This confirms that these nodes are already optimized for their scale.

2) Scenario B: Results for $k = 5$: The granular partition targets Cluster 3, which accurately isolates the most dysfunctional nodes in the network (40 airports with an average delay of 29.9 min).

- **Operational Scope:** The engine explores alternatives for smaller, "provincial" airports (e.g., MGW, VEL, ROW) with capacity often below 10 connections.
- **Efficiency:** Because these nodes have lower capacity requirements, the LSH-triggered search identifies a massive pool of **2,176 candidates**.
- **Impact:** After applying the Jaccard filter (> 0.4), the system yields 157 high-quality recommendations with a significant average gain of **13.8 minutes** (more than double that of Scenario A). In extreme cases (e.g., MGW to JST), the potential gain exceeds 60 minutes.

3) Conclusion: Strategic Trade-off: The comparison demonstrates a fundamental insight for network management:

- **In** $k = 2$, we obtain **High-Stakes** recommendations: switching between international hubs involves massive passenger volumes, making even a 6-minute gain strategically significant despite the low percentage improvement.
- **In** $k = 5$, we obtain **High-Efficiency** recommendations: the system identifies systemic failures in regional nodes where the gain is substantial, but the impact is localized to fewer passengers.

This dual analysis proves that while the "Fragility Paradox" is absent in the global backbone, it is a dominant reality in the operational periphery, where the recommendation system provides its most dramatic results.

VI. Conclusion

This study integrates structural topology and operational performance to investigate systemic fragility in the global aviation network. Our multi-phase analysis reveals that:

- **Structural dominance does not imply operational vulnerability:** Highly central hubs exhibit stable and predictable delays, contradicting the traditional *Fragility Paradox*.
- **Operational inefficiencies are peripheral:** The true bottlenecks reside in low-degree, low-centrality airports, which suffer from high average delays and variance.

- **Clustering and profiling are effective diagnostic tools:** Hierarchical, K-Means, and DBSCAN analyses consistently isolate peripheral nodes as critical inefficiency points, while revealing functional patterns among well-performing hubs.
- **Scalable recommendation systems are feasible:** The ”Smart Switch” engine demonstrates that candidate alternatives can be identified efficiently using LSH, even though exhaustive search is feasible for small datasets. Parameter tuning favors high recall to ensure no valid alternatives are missed, while speedup is already significant.

Overall, our results highlight the importance of combining topological and operational perspectives. Interventions aimed at improving network resilience should focus on peripheral inefficiencies rather than central hubs. Moreover, the proposed prescriptive framework provides a scalable approach to identify operationally robust alternatives, offering practical guidance for aviation management and planning.

Future work will focus on expanding the dataset to include a full annual record of flights, enabling the analysis of temporal trends and seasonal patterns. This will allow a more comprehensive evaluation of operational dynamics over time and support data-driven strategies for long-term resilience in global aviation networks.